

Unit 0: Workshop Setup & Environment

Dr Christian Engels
christian_engels@outlook.com

Applied Timeseries in R
7 Feb 2026

Certificate in Engagement with Practice in Finance (CEPF)
Durham University Business School



Today's Programme

Time	Unit	Topic
09:30 – 10:00	0	Setup: GitHub, Codespaces, R environment
10:00 – 10:45	1	Time Series & Tidyverse Foundations
10:45 – 11:00		<i>Break</i>
11:00 – 12:00	2	Random Walks, Stationarity & Unit Root Tests
12:00 – 13:00		<i>Lunch</i>
13:00 – 14:00	3	ARMA/ARIMA Models & Forecasting
14:00 – 14:15		<i>Break</i>
14:15 – 15:15	4	Volatility Models (ARCH/GARCH)
15:15 – 15:30		<i>Break</i>
15:30 – 16:15	5	Portfolio Optimisation & Risk Management
16:15 – 16:30	6	Wrap-up & Resources

You will be able to:

- 1 Download, wrangle and visualise financial time series in R
- 2 Test for stationarity and interpret ACF/PACF plots
- 3 Fit and forecast ARIMA models for the conditional mean
- 4 Specify, estimate and compare GARCH models for volatility
- 5 Construct an optimised portfolio and backtest its Value at Risk

Each unit is delivered as a pair:

- **Slide deck** — theory, equations, key ideas
- **Companion R script** — code you run alongside (named `Unit-N-Topic.R`, e.g. `Unit-1-TimeSeries-Foundations.R`)

Hands-On Workshop

You will write and run R code throughout the day. Experiment, break things, ask questions!

Step 1 — Create a GitHub Account

- 1 Go to <https://github.com>
- 2 Click **Sign up**
- 3 Follow the on-screen prompts to enter your email, password and username
- 4 Complete the verification puzzle

Already registered?

Sign in at <https://github.com/login> and skip to Step 2.

Step 2 — Accept the GitHub Classroom Assignment

- 1 Open the assignment link:

`classroom.github.com/a/lpdeYbn_`

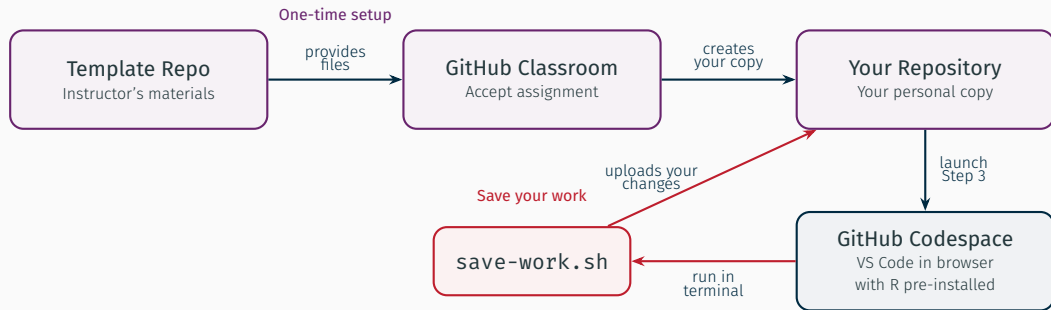
- 2 Click **Accept this assignment** — GitHub will create your own copy of the workshop repository
- 3 Once ready, find your repository under:

`github.com/orgs/Applied-Timeseries-in-R/repositories`

Your personal copy

All slides, R scripts and data files live in your repository. You will launch GitHub Codespaces from there in the next step.

How It All Fits Together



Repository

A project folder on GitHub that keeps a full history of every change.

Codespace

A complete computer in your browser — R and all packages pre-installed.

save-work.sh

Run this in the terminal to save your code back to your repository.

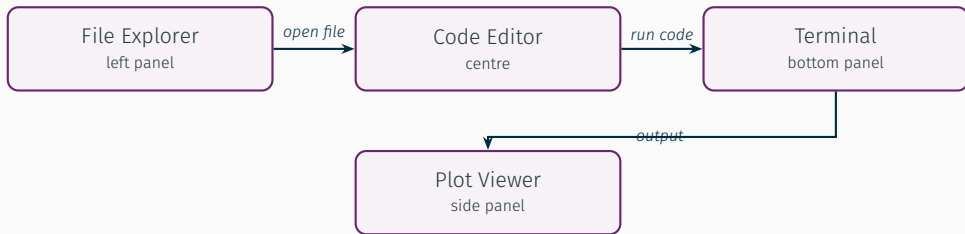
Step 3 — Launch a Codespace

- 1 On the repository page, click the green `<> Code` button
- 2 Switch to the **Codespaces** tab
- 3 Click **Create codespace on main**
- 4 Wait for the container to build — a VS Code editor will appear in your browser

First launch takes 2–5 minutes

The Codespace installs R, all packages and the workshop files automatically. Subsequent launches are faster. No local installation is required.

Navigating Your Codespace



- Open any `.R` file from the file explorer on the left
- Run it in the terminal: `Rscript Unit-0-Setup.R`

In the terminal, type:

```
R --version
```

You should see **R version 4.3** or later.

Packages We Use Today

Data & wrangling	tidyverse, tidyfinance, scales, broom
Time series	tsibble, fable, feasts, fabletools
Econometrics	fixest, tseries, moments
Volatility	rugarch, forecast
Portfolios	PortfolioAnalytics, xts, zoo

Step 5 — Run the Setup Script

- 1 In the terminal, run: `Rscript Unit-0-Setup.R`
- 2 The script installs any missing packages, loads every library and runs a quick smoke test.
- 3 You will see package-by-package status messages, followed by:

```
=====  
All packages loaded successfully!  
You are ready for the workshop.  
=====
```

Common issues:

- Package install timeout → re-run the script
- Compilation error for a package → ask the instructor
- Codespace won't start → refresh the browser page

Getting help:

- Raise your hand or post in the chat
- Read the error message in the terminal carefully
- In R, `?function_name` opens the help page

Ready to Begin!

You have: a GitHub account, a running Codespace and a verified R environment.

Unit 1: Time Series & Tidyverse Foundations →